

BIOTECHNOLOGY: DNA EXTRACTION

Grade 6 STEM Lesson — Discover the Tiny World Inside You!

■ What is Biotechnology?

Biotechnology means using **living things** to solve problems and make useful products! Scientists use tiny living organisms to create amazing solutions across many fields.

■ Medicine	Making medicines to help sick people
■ Agriculture	Improving crops so farmers can feed more people
■ Environment	Cleaning up pollution using natural organisms

■ Meet DNA — The Building Blocks of Life

DNA (Deoxyribonucleic Acid) carries the instructions for how all living things grow and function. Think of it like a **recipe book inside every cell** that tells your body how to build and run itself!

■ Fun Facts About DNA:

- 01 DNA is shaped like a **twisted ladder** called a *double helix*
- 02 Humans share about **99.9%** of their DNA with each other
- 03 DNA is found in **almost every cell** of your body

■ Why Extract DNA?

Scientists extract DNA for many important reasons:

- 01 **Study Genes & Traits**
Examine the genes and traits that make every living thing unique
- 02 **Forensic Science**
Solve crimes by analysing DNA evidence left at the scene
- 03 **Better Crops & Medicine**
Improve crops for farmers and develop new medicines for people

■ Materials Needed for DNA Extraction

#	Material	Purpose
01	Strawberries or Bananas	Source of DNA — fruit cells are easy to break open
02	Soap or Liquid Detergent	Breaks open cell walls to release the DNA
03	Table Salt	Helps clump the DNA strands together
04	Water	Used to create the extraction solution
05	Coffee Filter or Cloth	Strains out solid fruit bits from the liquid
06	Chilled Rubbing Alcohol / Ethanol	Makes DNA precipitate (appear) as white threads
07	Clear Container or Test Tube	To collect and observe the extracted DNA
08	Fork or Spoon	For mashing the fruit

■■ Step-by-Step Procedure

1

Step 1 — Prepare the Fruit Sample

Mash strawberries or bananas gently using a fork until smooth. This physical action breaks open the cells and begins releasing the DNA from inside.

2

Step 2 — Add the Soap Solution

Mix 1 teaspoon of liquid soap, a pinch of salt, and 100 ml of water in a bowl. Pour this over the mashed fruit and stir gently. The soap dissolves the cell membrane, while the salt helps the DNA clump together.

3

Step 3 — Filter the Mixture

Carefully pour the mixture through a coffee filter or cloth into a clear container. This removes the solid fruit pieces, leaving a clear liquid that contains the DNA.

4

Step 4 — Add Cold Alcohol

Slowly and gently pour chilled rubbing alcohol or ethanol on top of the filtered liquid. Tilt the container so the alcohol flows down the side. DNA is not soluble in alcohol, so it rises to the top as white, stringy threads.

5

Step 5 — Observe the DNA

White, stringy DNA strands will appear at the boundary between the two liquid layers! Use a wooden stick or glass rod to gently lift the strands out and observe them. Congratulations — you can now see the actual instructions that make living things work!

■ The Science Behind It

What You Used	Why It Works
Mashing the Fruit	Physical force breaks open the cell walls
Soap / Detergent	Dissolves the fatty cell membrane surrounding the nucleus
Salt	Neutralises the negative charge on DNA, helping strands clump
Filtering	Removes large solid particles so only liquid (with DNA) passes through
Cold Alcohol	DNA is insoluble in alcohol — it precipitates out as visible strands

■ Real-Life Applications of DNA Extraction

- **Forensic Science**
Detectives use DNA extracted from crime scenes to identify suspects or prove innocence.
- **Agriculture**
Scientists improve crop varieties by understanding the DNA of plants, making them stronger and more nutritious.
- **Medicine**
Researchers develop new treatments and cures by studying the genes linked to diseases.
- **Genetic Conditions**
Doctors can diagnose inherited conditions by analysing a patient's DNA.

■■ Safety Tips

- Always ask an adult before handling chemicals such as rubbing alcohol.
- Use clean equipment to ensure accurate and uncontaminated results.
- Wash your hands thoroughly with soap and water after the experiment.
- Work carefully and slowly throughout the process — do not rush.
- Keep alcohol away from open flames as it is flammable.

■ Summary — What We Learned Today

- Biotechnology uses living things to solve real-world problems.
- DNA is the instruction book found inside almost every living cell.
- DNA extraction separates DNA from other cell parts so we can study it.
- Simple household materials — soap, salt, and cold alcohol — can extract visible DNA.
- DNA extraction has powerful applications in medicine, farming, and forensics.

■ Keep asking questions — every great scientist started just like you! ■